

6 (a) Fig. 6.1 shows the energy levels of sodium gas.

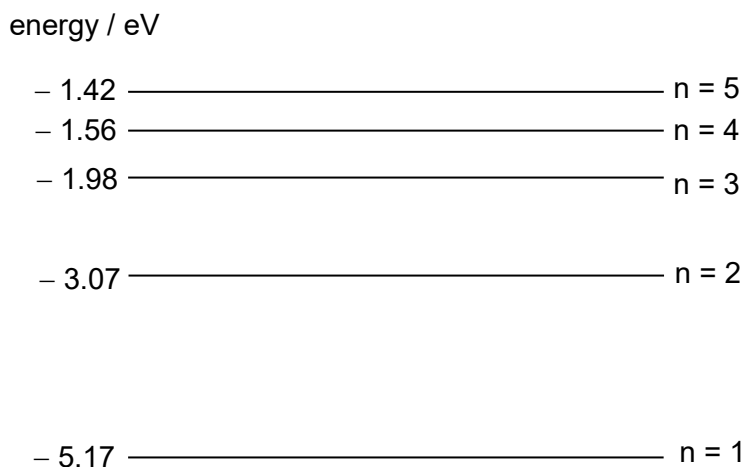


Fig. 6.1

The cool sodium gas at low pressure is bombarded by incoming electrons with kinetic energy of 3.70 eV.

(i) On Fig. 6.1, draw arrows to show all the possible de-excitation transitions that will be observed if all the sodium atoms are at ground state initially. [2]

(ii) Fig. 6.2 shows the emission line due to the transition from $n = 2$ to $n = 1$.

On Fig. 6.2, sketch the positions of all the other possible emission lines observed.

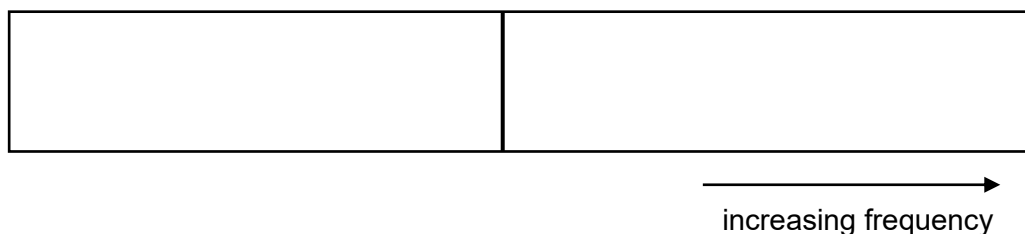


Fig. 6.2

[2]

(iii) The cool sodium gas is now bombarded by incoming photons of energy of 3.70 eV instead.

State and explain the number of emission lines observed, if any.

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 [1]

- (b) Fig. 6.3 shows the variation against wavelength of the relative intensity of emitted X-rays radiation from an X-ray tube. In the evacuated X-ray tube, electrons emitted at the cathode are accelerated from rest using an accelerating voltage to hit a target made of a metal at the anode and produce X-rays as a result.

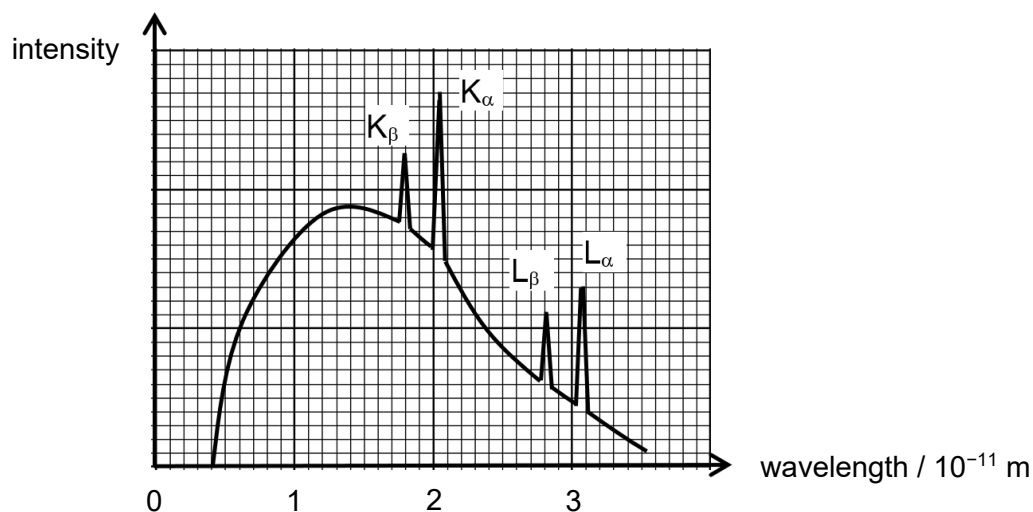


Fig. 6.3

- (i) Explain why the X-ray tube is evacuated.

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 [1]

- (ii) Explain what is meant by a *photon*.

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 [1]

- (iii) Determine the maximum energy of the X-ray photons emitted.

energy = J [2]

- (iv) Hence, show that the accelerating voltage required to accelerate the electrons in the tube is 310 kV.

[1]

7 Read the passage below and answer the questions that follow